# C MAN

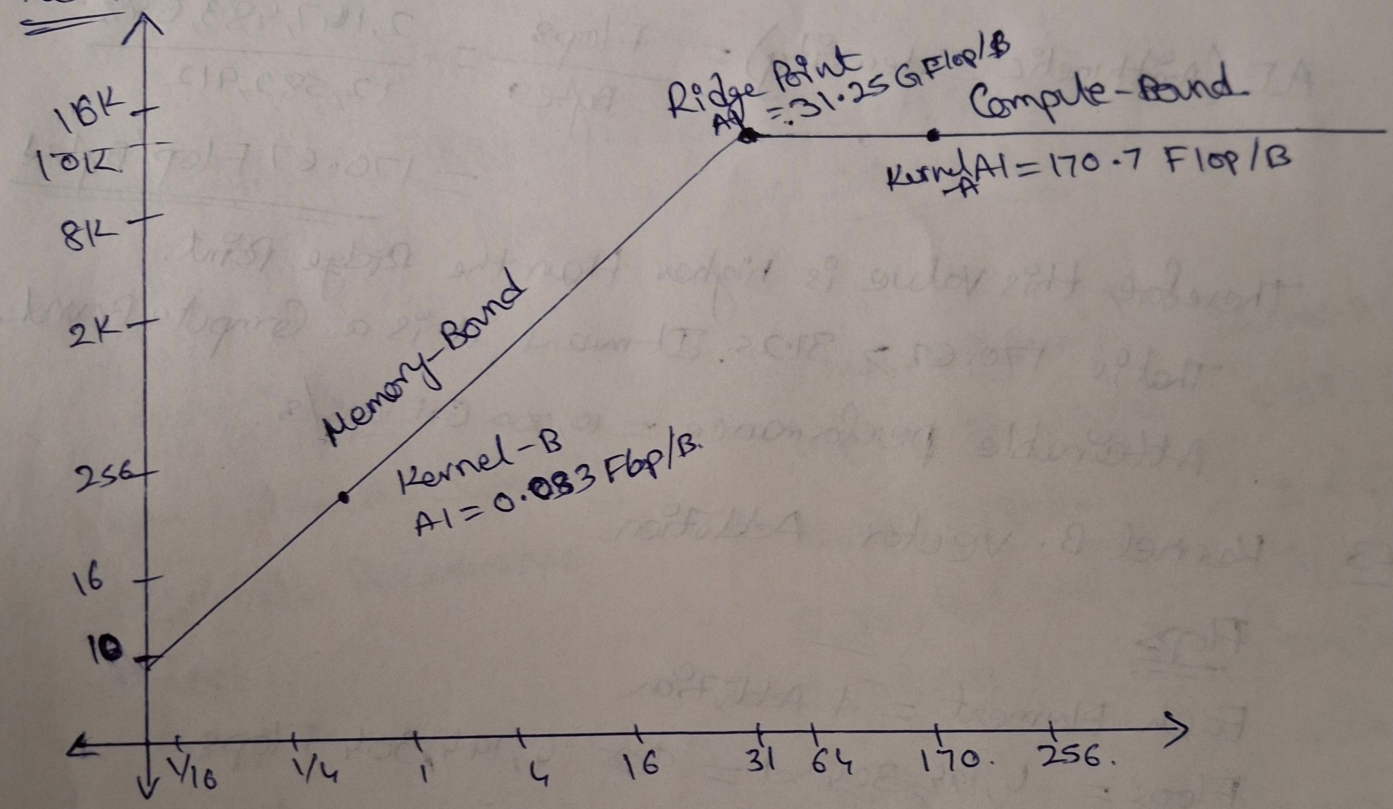
## Given Values

Peak Compute = 10 TFlops = 10,000 GFlop/s

Peak Bandwidth = 320 GB/s

Ridge Point = 10,000 ÷ 320 = 31.25 Flop/byte.

## Task1:




## Task2: Kernel A: Dense GEMM (1024x1024)

Flop's Calculation (N=1024)

$$\text{Flops} = 2 \times N^3$$
$$= 2 \times (1024)^3$$
$$= 2 \times 1{,}073{,}741{,}824$$
$$= 2{,}147{,}483{,}648 \text{ Flops}$$
$$= 2.147 \text{ GFlop}$$

Bytes: 1 Matrix = 1024 × 1024 × 4
= 4,194,304 Bytes

3 Matrices = 3 × 1 Matrix
= 3 × 4,194,304 bytes
Bytes = 12,582,912 bytes

AI (Arithmetic Intensity) = $\frac{\text{Flops}}{\text{Bytes}} = \frac{2,147,483,648}{12,582,912}$
= 170.67 Flop/Bytes

Therefore, this value is higher than the Ridge Point.
That is 170.67 > 31.25. It means it is a Compute Bound.
Attainable performance = 10,000 GFlops/s.

Task 3 Kernel B: Vector Addition.

Flops
Each Element = 1 Addition
Flops = 4,194,304 × 1 = 4,194,304 Flops
Flops = 0.004 GFlops

Bytes: (2 input and 1 output) (FP32)
1 Matrix = 1024 × 1024 × 4
= 4,194,304 Bytes
~~1 vector~~ 4,194,304 (1 Matrix)
1 Vector = 4,194,304 × (4) = 16,777,216 bytes
3 Vectors = (3) × 16,777,216
Bytes = 50,331,648 bytes = 50.33 MB

Arithmetic Intensity (AI) = Flops / Bytes

= 4,194,304 / 50,331,648

= 1/12

= 0.083 Flop/Byte

As this is less than the Ridge Point it is Memory Bound.

Attainable Performance = 0.083 × 320

= 26.67 GFlop/s

Task 4

| Analysis table | Kernel-A (GEMM) | Kernel-B (Vector ADD) |
|---|---|---|
| AI – Arithmetic Intensity | 170.67 Flop/Byte | 0.083 Flop/Byte |
| Attainable Performance | 10,000 GFlop/s | 26.67 GFlop/s |
| Flops | 2,147,483,648 | 4,194,304 |
| Bytes | 12,582,912 | 50,331,648 |
| Bound | Compute | Memory |
| Architectural (Recommendation): | In this Tensor Cores and MAC Units can be added. This also Raises Performance, More Compute Units, Wider SIMD | In this Switch to HBM, Wider BUS, Moving Data faster is the only lever. |